

EE 412/512 HW Assignment #3

Due November 12, 2025

Write the following statement on the first line of your solution sheet:

“I completed this homework assignment independently.”

The aim of this assignment is to design an amplifier on a microstrip substrate using ATF38143 transistor, whose S-parameters are provided in Touchstone format from 500 MHz to 18 GHz, in Moodle. In this format, the S-parameters and noise parameters are listed in a text file as a function of frequency. Choose one of the bias points of the transistor. Insert s2p icon in the schematic and choose the s2p files of the chosen bias. Connect input and output ports.

1. Plot Stability Factor (K) as a function of frequency from 500MHz to 10GHz as a rectangular plot (Insert StabFact icon in the schematic).
2. Plot the maximum available gain (MAG)/maximum stable gain (MSG) in dB from 500MHz to 10GHz as a rectangular plot (Insert MaxGain icon in the schematic). Note that when $K < 1$, we observe MSG, when $K > 1$, we observe MAG, and that there is a kink in the gain curve at the transition.
3. Plot the source and load stability circles on the unity Smith chart in the frequency range 0.5GHz to 10GHz. (Insert SStbCir and LStbCir icons found under Simulation-S_Param into the schematic. Uncheck Autoscale in Plot Options to set the unit circle.)
4. Terminate the input of the transistor with a source impedance value that is in the unstable region and show that the S_{22} goes outside the unity circle by plotting S_{22} .
5. Choose a frequency (f_1) for your amplifier. Stabilize the transistor at *all* frequencies by adding appropriate resistive network(s) at the input and/or output. You can use the tune option of ADS to find appropriate values while observing stability circles and the maximum gain plot. Try to maximize MAG in your chosen frequency as high as possible (close to the MSG or MAG values found earlier). Note that the gain will be maximized if $K(>1)$ is as close to unity as possible. Present your circuit, plot K, and MAG.
6. Set Start frequency to f_1 and Stop frequency to f_1 so that you have a simulation at just one frequency. Insert the GaCircle icon into the schematic and set the function in GaCircle to `ga_circle(S, , 2, 0.1)`. On the results window, insert a Smith chart, and draw Gacircle1. This will draw a dot (MAG) and a circle (MAG-0.1 in dB). The dot indicates the optimal source impedance values at f_1 , while the circle indicates the possible source impedance values at f_1 with a 0.1 dB gain reduction.

7. Insert your input matching network using lossless elements between the input port and the stabilized transistor. Choose your circuit topology such that the 50Ω source impedance can be transformed to the position of the dot. When the tuning network elements are inserted, the position of the dot changes. Tune the values of the input matching network such that the dot is at the center of the Smith chart (The best impedance value is now 50Ω .)
8. Insert the GlCircle icon in the schematic. This will help you design the output matching network. Set the function to `gl_circle(S,,2,0.2)`. This will draw one dot and one circle in the Smith chart. The dot indicates the best load impedance at f_1 , while the circle shows the possible load impedance values with a 0.2dB gain reduction.
9. Insert the output matching network using lossless elements between the output port and the stabilized transistor. Choose a circuit topology that transforms the 50Ω load impedance to the dot shown. (Equivalently, choose a circuit topology that transforms the complex conjugate of the dot to 50Ω .) Tune the values of the output matching network such that the dot coincides with the center. This will ensure that G_T equals G_A , and hence $|S_{21}|$ equals MaxGain at that frequency.
10. Plot G_T (S_{21}) in dB as a function of frequency (in a 20% fractional bandwidth).
11. Determine the frequency band where $|S_{11}|$ and $|S_{22}|$ of the amplifier are both less than -12 dB. Use markers to show the frequency band.
12. Find the noise figure of your amplifier for a source impedance of 50Ω . For this purpose, in the S-Parameter block, check Calculate noise in the Noise tab. Set Noise input port 1, Noise output port 2. Plot $nf(2)$ in a rectangular plot. Plot also NFmin. NFmin is the minimum noise figure that can be achieved.
13. Set the frequency to a single frequency of f_1 to eliminate a large number of circles. Insert NsCircle in the schematic and set the function to `NsCircle1=ns_circle(NFmin,Sopt,Rn/50,,3,0.2)`. Insert NsCircle1 into the same Smith chart that contains the G_A circles. This will draw three circles on the Smith chart corresponding to NFmin, NFmin+0.2, NFmin+0.4 dB at f_1 .
14. Noise figure optimization: If the noise circles can be brought closer to the center of the Smith chart, the noise figure will improve. When the input matching network is varied, the G_A circles will no longer be centered. Therefore, a compromise exists between maximum gain and minimum noise figure. Modify the input matching network so that the 0.2dB noise figure circle touches the

center while G_a is reduced as little as possible. This will reduce the gain, but the noise figure will be $NF_{min}+0.2$. Retune the output G_{circle} so that they are centered.

15. Plot S_{21} , S_{11} , and S_{22} magnitudes in dB. Plot the noise figure and F_{min} separately.
16. Insert a very small inductor between the source and ground of the transistor. If the source is already grounded, you need to use an S2P icon with an unconnected source pin. Using this inductor, it is possible to bring the noise figure and the G_a circles closer together. If the inductor is made larger, MAG will reduce. Since the two-port is modified, check that the stability factor is still above unity at all frequencies. Insertion of the small inductor may also help in stability, so you can modify the stability circuits to improve MAG. Retune your matching circuits to achieve a good noise figure, good S_{11} and S_{22} , and a somewhat reduced S_{21} .